Chapter 13, Problem 16PP

Problem

Calculate the turns ratio of an ideal transformer required to match a 8-Ω load to a source with internal impedance of 800 Ω. Find the load voltage when the source voltage is 300 V.

Step-by-step solution

Step 1 of 3

From the transformation ratio of a transformer,

$$V_1 = \frac{V_2}{n}$$
$$I_1 = nI_2$$

The input impedance as seen from the source is,

$$\begin{aligned} Z_{in} &= \frac{V_1}{I_1} \\ &= \frac{1}{n^2}\frac{V_2}{I_2} \\ &= \frac{Z_L}{n^2} \end{aligned}$$

$$\text{where } Z_L = \frac{V_2}{I_2}$$

Step 2 of 3

Calculate the value of turns ratio.

$$n^2 = \frac{Z_L}{Z_{in}}$$
$$n = \sqrt{\frac{Z_L}{Z_{in}}}$$

Substitute $8\ \Omega$ for Z_L and $800\ \Omega$ for Z_{in}.

$$\begin{aligned} n &= \sqrt{\frac{8}{800}} \\ &= \sqrt{\frac{1}{100}} \\ &= 0.1 \end{aligned}$$

Thus, the value of turns ratio is $\boxed{0.1}$.

Step 3 of 3

Calculate the value of load voltage, V_L .

$$\frac{V_L}{V_S} = n$$
$$V_L = nV_S$$

Substitute 0.1 for n and $300\ V$ for V_S.

$$\begin{aligned} V_L &= 0.1 \times 300 \\ &= 30\,V \end{aligned}$$

Thus, the value of the load voltage is $\boxed{30\,V}$.